Photosynthesis & limiting factors

Make & use glucose

1 The reaction

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Chlorophyll absorbs light; energy transfers from the environment to chloroplasts. **Endothermic.**

CO_2 enters stomata; roots/xylem supply water. Light is energy, not a material reactant.

2 Glucose pathways

Respiration releases energy; starch and fats/oils store resources. Cellulose strengthens cell walls.

Glucose and **nitrate ions** supply materials for amino acids → proteins.

Explain rate changes

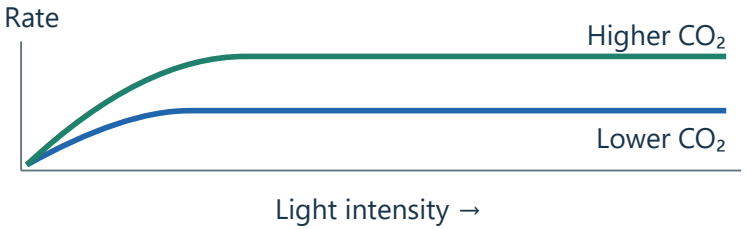
3 Four interacting factors

More light or CO_2 raises rate only while that factor limits it.

Warming towards optimum raises enzyme activity; excess heat can **denature enzymes.**

Less chlorophyll → less light absorption. Magnesium deficiency or disease can reduce chlorophyll.

4 Interpret the plateau



Rising rate with light → light is limiting. Plateau → another factor limits rate.

Compare curves at the same other conditions: if added CO_2 raises rate, CO_2 had limited it.

A plateau alone does not identify which factor limits rate.

Apply the maths

5 Inverse-square law

Intensity ∝ 1/distance². $I_2/I_1 = (d_1/d_2)^2$.

20 → 40 cm: intensity becomes **¼.** 30 → 15 cm: becomes **4×.**

Use consistent units. Photosynthesis rate need not change by the same factor.

6 Greenhouse decisions

Adding heat, light or CO_2 costs money. Improve the actual limiting factor.

Extra profit = extra crop income − extra cost. £240 − £170 = **£70.**

Highest photosynthesis rate is not always the most profitable setting.